

Flood Lines in Time: Temporality and Capitalization of Climate Risk in the Jersey Shore Housing Market

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Abstract

Despite significant environmental vulnerabilities, house prices on the Jersey Shore continue to rise. Although climate change exacerbates these vulnerabilities, primarily through rising sea levels and increased flooding from natural hazards, the value of homes and housing investment on the Jersey Shore is at an all-time high and continues to rise. The coastal nature of the area exacerbates the housing situation, as a high proportion of housing units are seasonal. This paper investigated the interconnection between climate risk and seasonality in residential property values on the Jersey Shore using a hedonic regression model with fixed effects. The findings reveal that, while properties in areas with a high proportion of seasonal homes command a premium, this is dependent on proximity to the ocean and reverses for inland properties after 2.6 miles from the ocean.

Keywords: coastal risk, flood zones, New Jersey, housing markets, climate capitalization

1 Introduction

Despite significant environmental vulnerabilities, house prices on the Jersey Shore continue to rise. Even though climate change exacerbates these vulnerabilities, primarily through rising sea levels and increased flooding from natural hazards, the value of homes on the Jersey Shore is at an all-time high and continues to rise. Moreover, despite little population growth, housing investment in the area continues to rise (Andrews, 2025). The current project aims to investigate this paradox. More specifically, I focus on how climate risk, defined as short- and long-term environmental vulnerabilities, is priced into property values. The commonsense assumption is that properties with high climate risk, for instance, located in a high-flooding risk zone, would receive a discount in property value. However, at the same time, having access to shore-related amenities, such as ocean view and beach proximity, would give them a premium, or in other words, increase

their value. Henceforth, the expected effect of being at the shore is ambiguous: some argue that properties in coastal communities, at least in the United States, are overvalued (Gourevitch et al., 2023; Hino & Burke, 2020). In contrast, others suggest that the climate risk is already priced into housing through insurance and other mechanisms (Collier & Cox, 2021; Cox et al., 2025).

Having an ocean and a beach as amenities also leads to certain characteristics of urban settlement, namely being a coastal community with high tourist attractions. At the Jersey Shore, as in other coastal communities, a significant share of housing is seasonal, meaning it is either a second home for individuals who live elsewhere or a short-term rental that people occupy while visiting the shore. Increasing the share of such housing utilization tends to distort the local economy, both in terms of housing availability for year-round residents and, more broadly, by relying on cheaper, service-oriented labor in tourist areas (Wassmer, 2022).

This paper, thus, focuses on the interaction between these two phenomena. More precisely, this study is concerned with whether there is a difference in climate risk price adjustment (which can be both a discount and a premium) for seasonal housing (i.e., second homes or short-term rentals) versus year-round housing (i.e., properties where residents reside for most of the year). To answer that question, the hedonic regression analysis is conducted with time and entity fixed effects on the sales transaction data in order to examine potential interaction between the seasonal character of the housing market at the Jersey Shore and climate risk capitalization into the property value using related climate proxies, distance to the ocean, and flood risk zone.

This study is structured as follows. First, I conduct a brief literature review of climate risk capitalization in housing scholarship and formulate the research question that drives this paper. Then, I described the main data sources, namely the tax assessor historical MOD-IV database. Then, describe the model specification, the dependent variable (sale price), the independent variables of interest (flood zone, distance to the shore, and share of seasonal housing units), and the control variables used. After that, I present and discuss the regression model results and point out potential limitations of the data-generating process, as well as issues with the research design. Lastly, I conclude the paper by outlining the potential policy implications of the main findings for the Jersey Shore housing market.

2 Literature Review

Literature on climate risk capitalization in housing markets has grown significantly in recent years, as flooding and sea-level rise become a bigger concern for property owners. As housing is usually the main financial asset of the population and a major part of the wealth accumulation, scholarship has been attempting to evaluate whether climate affects the housing price, as well as the extent to which it does. Overall, there is consensus on the presence of the aforementioned effect: properties located in flood zones or areas of possible sea-level rise receive a discount when sold or assessed. However, researchers point out that this effect is heterogeneous and depends on a variety of factors of the locality (Baldauf et al., 2020; Murfin & Spiegel, 2020; Sussman et al., 2014).

Gourevitch et al. (2023) find that residential properties exposed to the flood risk are over-

valued by around \$130 billion, especially those located in coastal areas with no laws regarding disclosure of flood risk. Another estimate by Hino and Burke (2020) provides a value of \$34 billion for floodplain homes overvaluation, while pointing out that this effect is more important for commercial property rather than residential. In contrast, Putra (2017) highlights that the presence of residential properties in high-risk flood zones is increasing, especially those previously affected by natural disasters, based on a case study of the housing submarket in the state of New Jersey. Gibson et al. (2017) suggests that each flood risk event, such as a hurricane or the publication of a new flood risk map, decreases the sale price of property by 5 percent. Similarly, other studies find that rather than climate risk affecting property values by itself, the more important aspect is whether information about climate risk exists in the housing market, both in terms of public awareness, available data due to laws and regulations, news, or whether community beliefs in climate change (Baldauf et al., 2020; Bunten & Kahn, 2014; Chen, 2025; Hino & Burke, 2020).

As climate risk capitalization in property values is a heterogeneous effect, the interest of the current paper lies in whether this effect differs between seasonal and year-round residential properties. Second home tourism and short-term rental properties have been shown to have a significant influence on the housing submarkets and local economies (Collado-González et al., 2025; Gil, 2024; Perles-Ribes et al., 2025; Valenzuela-Martin et al., 2025; Wassmer, 2022). Despite a considerable amount of scholarship focused on climate risk capitalization in property values and residential properties in particular, there is very limited coverage of how climate risk capitalization is different based on the seasonality status of the property, or in other words, whether there is any significant interaction between the seasonality status of the property and climate risk capitalization. Consequently, the research question that guides this paper could be formulated as follows: Is there a difference in climate risk price adjustment for seasonal (i.e., second homes or short-term rentals) versus year-round (i.e., properties where residents reside for most of the year)?

Climate risk capitalization increases the vulnerability of low-income communities (Gourevitch et al., 2023). Additionally, the rise of second homes and short-term rentals in housing markets can exacerbate the housing crisis (Collado-González et al., 2025; Gottlieb, 2013; Perles-Ribes et al., 2025), disproportionately impacting low-income communities. Consequently, the interplay between seasonality and climate risk is a significant policy concern. In areas like the Jersey Shore, where both factors are present in the housing market, they could further entrench existing inequalities and lead to the displacement of residents.

3 Data and Methods

3.1 Jersey Shore as a Study Area

This study specifically focuses on the Jersey Shore housing submarket, which has a significant risk of climate risk capitalization, as highlighted by previous studies (Andrews, 2025; Putra, 2017). The study area is defined as the coastal planning area defined in the New Jersey Coastal Area Facility Review Act of 1974, which authorized the Department of Environmental Protection

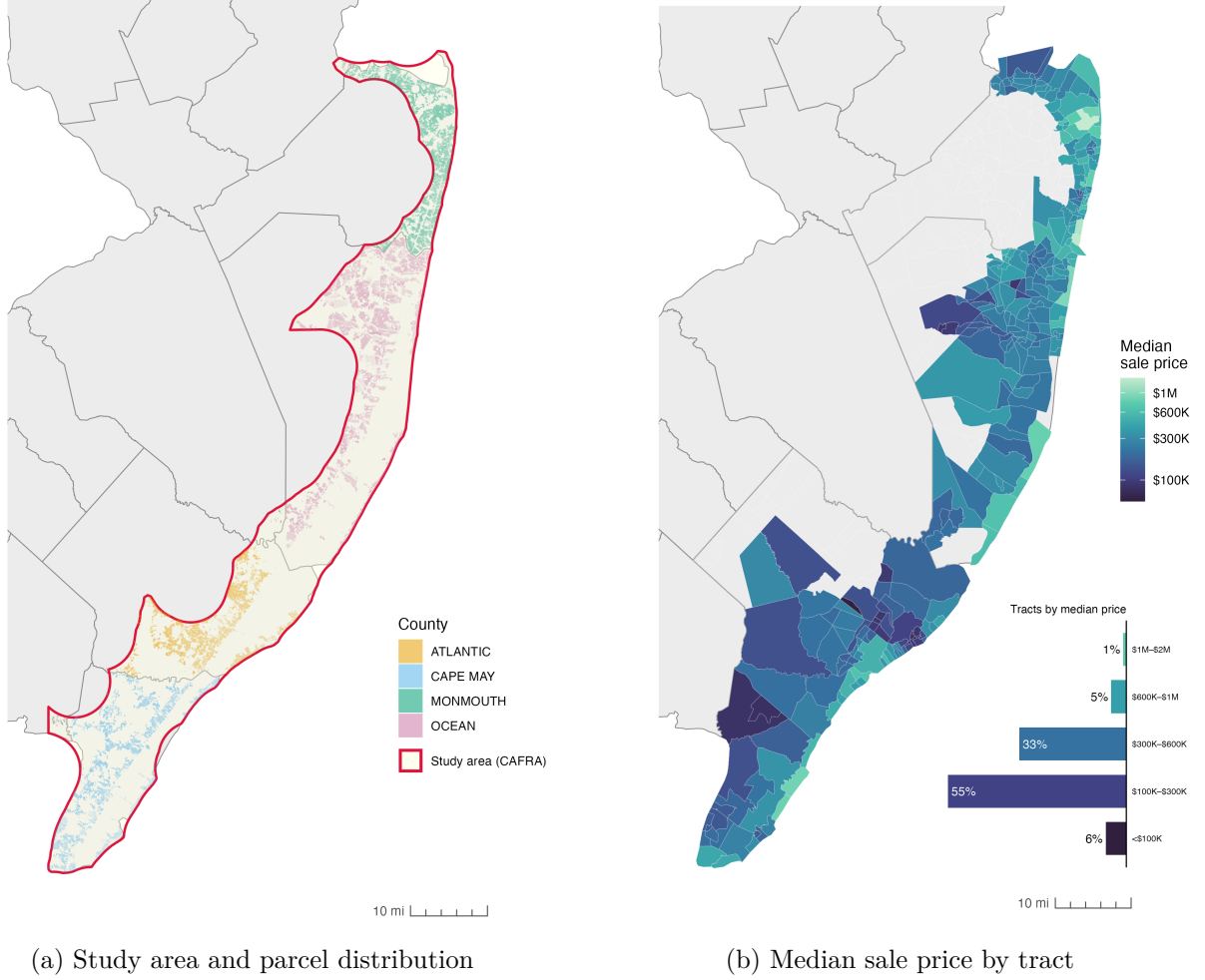


Figure 1: *Spatial context and key variable used in the analysis.* Note. Figure (a) shows the Jersey Shore study area and parcel sample by county; figure (b) maps census tract-level median sale price (2023).

to manage development along the coastline and is thus a coastal planning area. Simplified boundary of the area is presented in Figure 1a. It consists of the part of four counties: Atlantic, Cape May, Monmouth, and Ocean. The data collection and analysis described in the next sections are performed for this area.

3.2 Property Data

As the base dataset, this paper utilizes the New Jersey MOD-IV historical database. It contains real estate parcel data maintained by municipal assessors and includes information on parcel characteristics, including land and dwelling assessed values, building characteristics, and property taxes. Specifically, I collect data on residential parcels defined by property tax classes over 10 years, from 2015 to 2025. To enrich the collected data with additional sources and select only parcels located at the Jersey Shore, the parcel data is joined with spatial data on parcel

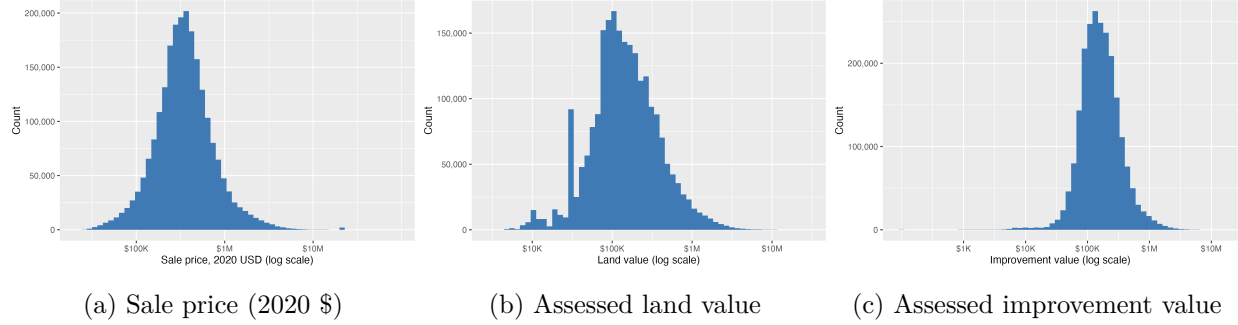


Figure 2: *Distributions of sale price and assessed values.* Note. Sale price is inflation-adjusted to 2020 dollars and excludes non-market transactions below \$20,000; land and improvement values are as assessed by the municipality.

geometries available on the NJ GIS website.

3.3 Additional Datasources

As a proxy for flood risk, I collect the Federal Emergency Management Agency’s (FEMA) National Flood Hazard Layer (NFHL) for the state of NJ. It consists of information on spatially referenced flood risk zones with different levels of risk. As a proxy for the seasonality status of the parcel, I downloaded American Community Survey (ACS) data on the number of seasonal housing units and the total number of housing units at the census tract level for two time periods (2013 and 2023). The US Census Bureau defines seasonal housing units as “seasonal housing units are those intended for occupancy only during certain seasons of the year and are found primarily in resort areas” (US Census Bureau, n.d.), i.e., units that are occupied by somebody with usual residence elsewhere only during particular times, which fits into the essence of second homes and short-term rental market at the Shore.

3.4 Regression Analysis

To examine the climate risk capitalization at the Jersey Shore’s housing submarket, this study employs a hedonic regression approach for pooled cross-sectional data. Hedonic regression has been a conventionally accepted model for studies where the dependent variable (DV) is related to the price of goods or services, including housing prices. Such an approach has also been employed in previous studies of climate risk capitalization (see, for instance, Putra, 2017). As described below, DV of the current study is sales price, which is regressed against variables of interest related to flooding risk, housing seasonality, and selected controls, making such research design applicable to the aforementioned research question. I describe model specification, included variables, their cleaning, and transformation, as well as research design choices in the following sections.

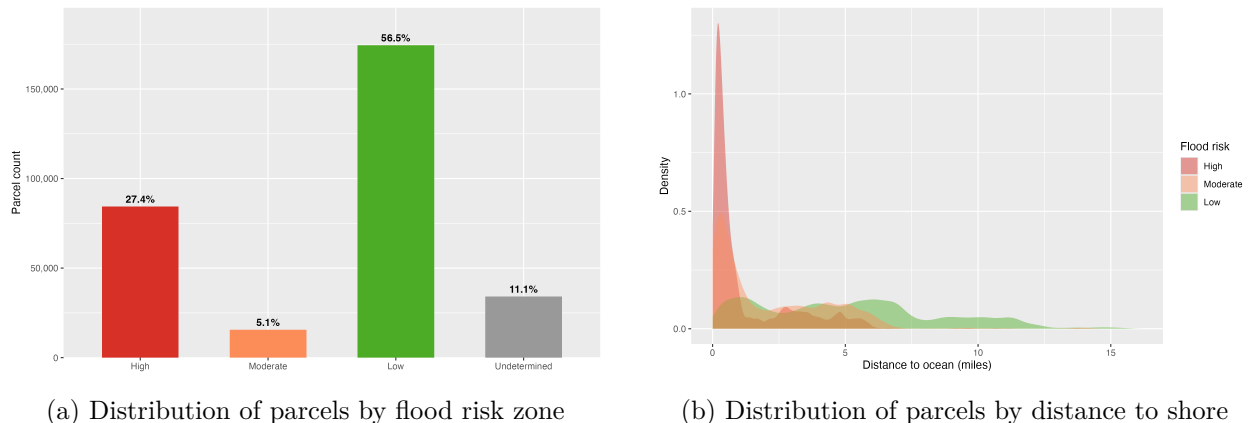


Figure 3: *Flood risk and proximity to shore for parcels in the study area.* Note. Figure (a) shows the share of parcels by FEMA flood risk zone; figure (b) shows the distribution of parcel distances to the shoreline.

3.4.1 Dependent Variable

The dependent variable in the regression model is the parcel’s sales price. The MOD-IV historical database includes information on the sale price and sale year, i.e., when this parcel of land was sold and for what dollar value. Each parcel can have an unlimited number of transaction events that would be reflected in the database. However, based on the conducted explanatory data analysis, very few parcels in the sample have had multiple transactions. Most of them have one or two record transactions. It is important to note that MOD-IV collects sales price information for the last sale that has occurred to the property, which means that if the parcel were last sold in the year 2000, that would be the recorded sale price that we utilize as DV.

To account for this aspect, we adjust the sales price from nominal dollars to real dollars in 2020 dollars, using Consumer Price Index data and the sales year variable provided in the MOD-IV database. The year 2020 is selected as the reference year to avoid the effects of the post-COVID-19 the post-COVID-19 rise in housing prices.

Additionally, all nominal sales are excluded from the analysis, meaning transactions that did not occur under market conditions but were rather transfers of property rights within the family or other related parties. These transactions usually are noted by the sale price of \$0 or \$1. Additionally, I filter out records where the sales price is less than \$20,000 for the same reason, that they are not real market-rate sale transactions. Scholarship conventions determined the value threshold. The dependent variable is log-transformed to address the right skew in the sales price data.

Figure 1b shows the median sale price aggregated at the census tract level. More than half of the tracts have a median selling price within the range of \$100,000 to \$300,000. The spatial pattern of median price is evident as well: tracts closer to the ocean have, on average, higher selling prices than inland tracts. This observation is consistent with other studies on the Shore and can

be explained by the fact that market actors value proximity to the ocean, including the beach and ocean views.

3.4.2 Independent Variables

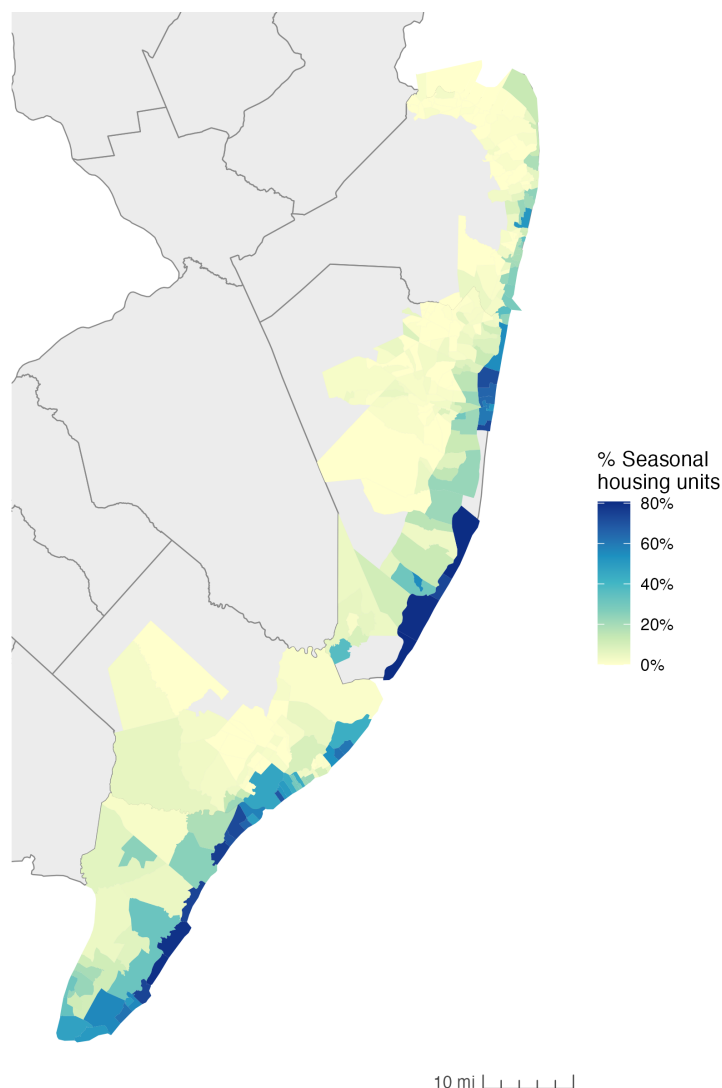


Figure 4: *Share of seasonal housing units by census tract.* Note. Percent of seasonal units derived from ACS data; tracts closer to the ocean tend to have a higher share of seasonal units.

There are three independent variables (IV) of interest. First, flood zone, a categorical variable generated from the FEMA NFHL data that classifies the floodplain into multiple zones. For simplification, those zones are aggregated into four types: high, moderate, and low risk, as well as an undetermined zone, which represents open water areas of the shore and areas where flooding risk exists but was not assessed. All parcels in the undetermined zones are excluded from the analysis, as it is not feasible to compare this category with zones with quantified flood

risk meaningfully. Low-risk served as the omitted reference category in the regression analysis. As shown in Figure 3a, the majority of parcels in our study area have a low flood risk. The second-most populous category is high risk, while very few parcels have a moderate risk of flooding.

Second, IV of interest is the distance to the shore, generated using spatial analysis techniques. Scholarship on climate risk capitalization has shown it to be an important variable (Murfin & Spiegel, 2020; Putra, 2017), which, however, might have unexpected effects. On the one hand, being close to the shore increases the risk of flooding during natural hazards and of long-term damage from sea-level rise. On the other hand, being close to the shore, especially having a view of the water or being within walking distance of the beach, is considered an advantage for the residential property. As evident from the Figure 3b, a major part of parcels in the sample are in proximity to the ocean, which also correlates with having a high level of flood risk. This variable is also log-transformed to account for the effect that distance has on property values: the difference between being 200 and 1000 feet away is assumed to be significant for the property values, while being 5 or 5.1 miles away is not supposed to have any effect.

Third, the seasonality variable, defined as the proxy for the percent of seasonal units in the census tract and generated from ACS data on the number of seasonal and total housing units in the tract. The proxy is calculated as

$$\text{Percent of seasonal units}_i = \frac{\text{Seasonal units}_i}{\text{Total units}_i}, \quad \forall i \in \{\text{census tracts}\} \quad (1)$$

It is important to note the potential limitation of such a proxy variable. As our original data are at the parcel level, while the seasonality variable is at the census tract level, this mismatch created multiple challenges for the research design, including the accuracy of measurement and the inclusion of fixed effects at the census tract level in the model. I describe these limitations and potential solutions in subsection 5.2. Figure 4 demonstrates the spatial distribution of seasonality units at the shore. Census tracts near the shore have a higher share of seasonal housing units, which is expected, as those areas are considered tourist areas. Surprisingly, the north of the shore has, on average, a lower share of seasonal units than the middle and southern parts.

Lastly, as this study is concerned with the interaction of seasonality and climate risk, I also introduce two interaction terms into the model: the interaction between distance to shore and percent of seasonal units, and the interaction between flood zone and percent of seasonal units, which serve as the main variables of interest in accordance with the research question.

3.4.3 Control Variables

Additionally, our model specification includes control variables, namely the age of the building, which has been shown to affect the sales price, and the assessed value of the land (parcel) and the assessed value of the improvements (building on the parcel). Whilst classical hedonic regression analysis studies generally take building characteristics into account, such as material, number of floors, presence of a garage, and square footage, the MOD-IV database is of substandard quality, and the building description variable is imbalanced. Utilizing the MOD-IV database would there-

fore result in a substantial reduction in our sample size. Because of that, I control for land and improvement value assessed by the municipality, which should serve as a reasonable control for the market sale price, as it has a direct effect on it. Both variables are log-transformed because price distributions tend to be right-skewed, which makes them closer to normal distributions, as shown in Figure 2b and Figure 2c.

3.4.4 Final Model Specification

Table 1: Summary Statistics

Variable	<i>N</i>	Mean	SD	Min	Median	Max
Sale Price (2020 \$)	2,246,512	490,680	817,915	22,280	341,629	86,588,969
Distance to Ocean (mi)	2,246,512	3.883	3.491	0.002	3.299	16.160
Flood Risk: High	2,246,512	0.299	0.458	0.000	0.000	1.000
Flood Risk: Moderate	2,246,512	0.058	0.234	0.000	0.000	1.000
Flood Risk: Low	2,246,512	0.643	0.479	0.000	1.000	1.000
Pct. Seasonal Units	2,246,512	0.188	0.258	0.000	0.045	0.807
Age (years)	2,246,512	50.422	27.965	1.000	47.000	223.000
Land Value (\$)	2,246,512	235,487	353,870	3,000	135,000	27,385,500
Improvement Value (\$)	2,246,512	217,070	237,892	100	155,000	13,250,900

Note. Flood risk dummies sum to 1. Sale price is inflation-adjusted to 2020 dollars.

Table 1 provides summary statistics of the variables included in the model. The median price of sale in 2020 dollars is \$ 341,629. Mean price of sales is \$490,680 with a maximum value of more than 85 million dollars. A mean higher than the median highlights the distributional skew discussed earlier. The mean distance to the ocean is 3.9 miles with a maximum distance of 16 miles. The share of seasonal units ranges from 0 to 0.8, with a mean of 0.19, indicating that a few census tracts exhibit high seasonality. It also means that there are tracts where almost all the housing units are seasonal. Average age of the dwelling is 50 years, with an average assessed value of \$217,070. Average assessed value of the land is slightly higher at \$235,487.

After performing all transformations and data cleaning and excluding observations with missing values, our sample comprises 2,246,512 parcel-level pooled cross-sectional observations for years 2015 to 2025.

Property values tend to exhibit spatial dependence, meaning that properties with similar values are often located near one another because of shared neighborhood characteristics (Basile et al., 2014; see Can & Megbolugbe, 1997). Because our model does not directly control for all unobserved local attributes (e.g., amenities, microclimate, and school quality) that may influence sale prices, we include county-level entity fixed effects. Although finer fixed effects (e.g., census tract) would better account for unobserved heterogeneity, this is not feasible given the construction of our seasonality variable. Specifically, because seasonality is measured at the census-tract level,

adding census-tract fixed effects would absorb this variation and prevent estimation of its coefficient. Since the dataset is pooled cross-sectional data, we also include year fixed effects.

Final model is specified as follows:

$$\begin{aligned} \ln(\text{Sale_Price}_{it}) = & \beta_0 + \beta_1 \log(X_{\text{distance to shore}}) + \beta_2 X_{\text{seasonality}} + \beta_3 X_{\text{FEMA zone}} \\ & + \beta_4 (X_{\text{FEMA zone}} \times X_{\text{seasonality}}) + \beta_5 (\log(X_{\text{distance to shore}}) \times X_{\text{seasonality}}) + \gamma' \mathbf{C}_{it} + \alpha_i + \delta_t + u_{it} \end{aligned} \quad (2)$$

where $\ln(\text{Sale_Price}_{it})$ is the natural log of the inflation-adjusted sale price for parcel i in year t ; $X_{\text{distance to shore}}$ is the log-transformed distance from the parcel centroid to the nearest shoreline; $X_{\text{seasonality}}$ is the share of seasonal housing units in the census tract; $X_{\text{FEMA zone}}$ is a categorical indicator of flood risk zone (high, moderate, or low); $\gamma' \mathbf{C}_{it}$ is a vector of control variables including log-transformed assessed land value, log-transformed assessed improvement value, and building age; α_i denotes county fixed effects; δ_t denotes year fixed effects; and u_{it} is the idiosyncratic error term.

Additionally, the main model is compared with the ordinary least squares (OLS) model, the fixed time effects model, and the model with fixed time effects and fixed entity effects at the census tract level. Due to the sample size ($n = 2,246,512$), clustered standard errors are reported at the county or census tract level for the corresponding fixed-effects models.

4 Results

Results of the regression analysis are reported in Table 2.

First, the OLS results indicate that multiple variables of interest are significant at the conventional level. Distance to the ocean, which has a negative coefficient, meaning that a 1% increase in distance increases the sale price by 0.04%, holding other variables constant. Being in the moderate flood risk zone decreases the sale price by approximately 2.64% compared to the low flood risk zone; however, being in the high risk zone does not show a similar significant effect. The seasonality variable is significant and has a positive coefficient, indicating that a 10 percentage point increase in the share of seasonal units is associated with approximately a 1.9% increase in sale price, holding other variables constant. However, this main effect cannot be interpreted in isolation, because the model also includes the interaction $\log(\text{Distance}) \times \text{Pct. Seasonal}$ ($\hat{\beta} = -0.196$), the full marginal effect of seasonality on log sale price is $0.188 - 0.196 \times \log(\text{Distance})$. At close distances, the net effect is positive. However, it diminishes and eventually reverses as the distance to the shore increases, meaning the positive seasonality effect on price is concentrated in parcels nearest to the water. All of the interaction terms are significant, with $\log(\text{Distance}) \times \text{Pct. Seasonal}$ and $\text{High Risk} \times \text{Pct. Seasonal}$ having a negative effect on the sale price, while $\text{Moderate Risk} \times \text{Pct. Seasonal}$ surprisingly has a positive coefficient. The model's R^2 is 0.66, indicating that the model explains 66% of the variation in the data. However, as the study's sample is pooled cross-sectional data, we can assume that OLS models overestimate coefficients due to unobserved heterogeneity that affects sales price, such as neighborhood characteristics. We can also assume that the model underestimates standard errors due to sample size, thereby affecting inference.

Table 2: Hedonic Regression Results

	log(Sale Price, 2020\$)			
	OLS	Year FE	Year + County FE	Year + Tract FE
	(1)	(2)	(3)	(4)
Constant	3.676*** (0.0059)			
log(Distance to Ocean)	0.0412*** (0.0004)	0.0451 (0.0206)	0.0410* (0.0146)	-0.0637*** (0.0102)
Flood Risk: High	0.0012 (0.0013)	-0.0002 (0.0015)	-0.0068 (0.0079)	0.0005 (0.0134)
Flood Risk: Moderate	-0.0264*** (0.0018)	-0.0319 (0.0238)	-0.0363 (0.0239)	-0.0161 (0.0132)
Pct. Seasonal Units	0.1876*** (0.0045)	0.1955 (0.1007)	0.2261** (0.0593)	
Age (years)	-0.0016*** (1.22×10^{-5})	-0.0016** (0.0003)	-0.0016*** (0.0002)	-0.0012*** (0.0002)
log(Land Value)	0.3899*** (0.0005)	0.3994*** (0.0438)	0.4007*** (0.0451)	0.3366*** (0.0381)
log(Improvement Value)	0.3721*** (0.0005)	0.3686*** (0.0091)	0.3665*** (0.0115)	0.3544*** (0.0153)
log(Distance) $\times$ Pct. Seasonal	-0.1960*** (0.0013)	-0.1978*** (0.0227)	-0.1883*** (0.0122)	
High Risk $\times$ Pct. Seasonal	-0.0618*** (0.0052)	-0.0486 (0.1090)	-0.0495 (0.1024)	
Moderate Risk $\times$ Pct. Seasonal	0.0145** (0.0067)	0.0238 (0.1435)	0.0389 (0.1377)	
Standard-Errors	IID		County	Census Tract
Observations	2,246,512	2,246,512	2,246,512	2,246,512
R ²	0.66504	0.71732	0.71783	0.74351
Adjusted R ²	0.66504	0.71732	0.71782	0.74348
Year fixed effects		✓	✓	✓
County fixed effects			✓	
Census Tract fixed effects				✓

Note. Significance codes: * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$. Standard errors in parentheses.

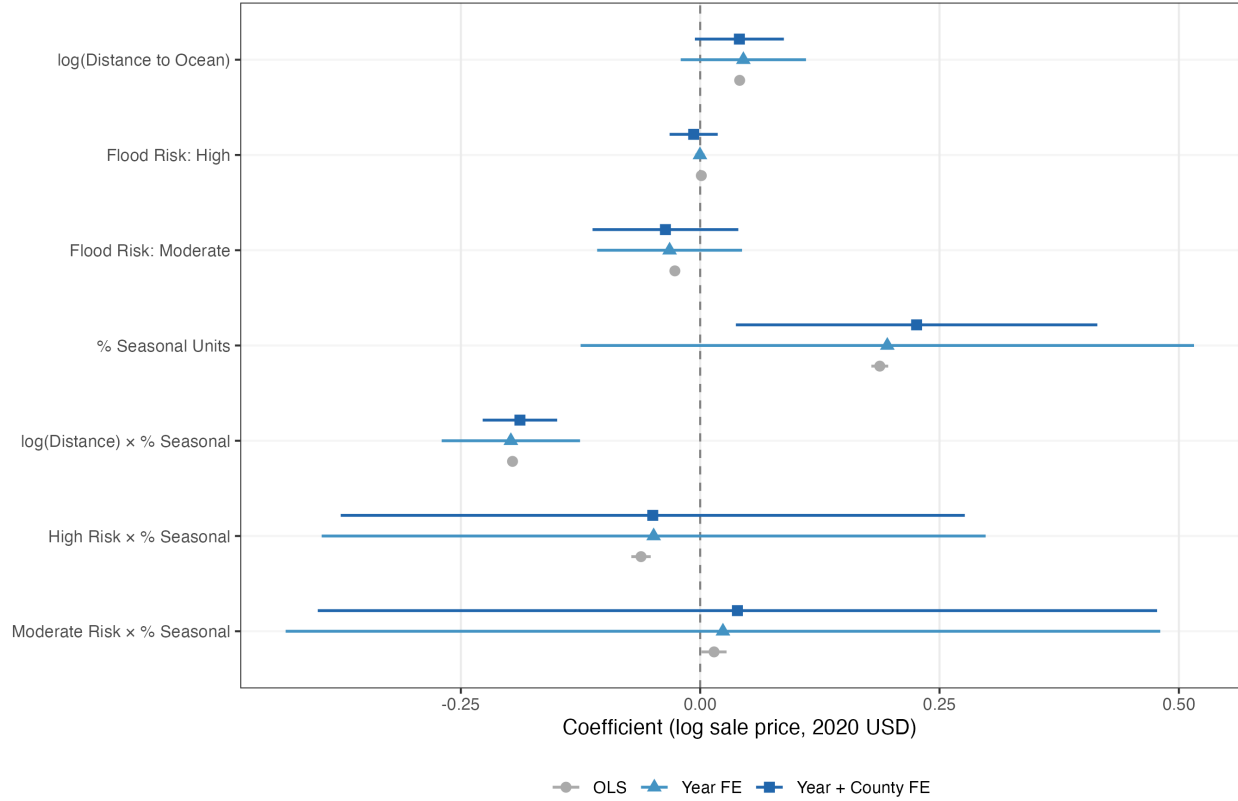


Figure 5: *Coefficient estimates for variables of interest across hedonic regression specifications.* *Note.* Points denote point estimates with 95% confidence intervals. Gray circles represent the baseline OLS model; blue triangles represent the model with year fixed effects; dark blue squares represent the model with year and county fixed effects. The vertical dashed line at zero serves as a reference for null effects. Wide confidence intervals for the interaction terms reflect the small number of clustering units ($n = 4$ counties) rather than model misspecification.

When the year fixed effects are introduced, R^2 increases to 0.71, signifying that time variation has significant explanatory power in the model. However, this also leads to most of the coefficients for the variables of interest losing significance because standard errors are clustered at the county level. However, interaction between distance and seasonality remains significant at the conventional level. Moreover, the magnitude of the coefficient changes only slightly, indicating the model's correct specification and the importance of these variables.

When the county-level fixed effects are added in addition to the year fixed effects, R^2 marginally increases, indicating that county-level fixed effects have very little explanatory power. However, some coefficients, namely distance to the shore and percent of seasonal units, become significant, indicating that the model's 2 standard errors were inflated by between-county confounding. Importantly, most of the coefficients remain stable, indicating the correct specification of the model. The only exception is the coefficient for the seasonality variable, which increases from 0.1955 to 0.2261.

Lastly, when the standard errors are clustered at the census block level, R^2 increases from 0.71 to 0.74, indicating a better fit for the model than for model 3. It also means that census tracts better account for unobserved heterogeneity in the data. Moreover, the distance-to-shore coefficient gains significance, but the direction of the effect reverses. This is explained by the fact that the effect previously captured by the interaction term between seasonality and distance to shore has moved to the distance-to-shore variable. It is also important to note that all control variables remain significant across all 4 models and do not exhibit large changes in coefficients from model to model, indicating correct model specification.

Despite most of the coefficients for the IVs of interest not being statistically significant at the conventional level, there is an argument that this is due to the clustered standard error specification rather than model misspecification or the lack of association between the variables. When the standard errors are clustered at the county level, SEs are inflated due to the small number of entities ($n=4$). This is further evidenced by the fact that introducing fixed effects barely changes the magnitude of the coefficients. Figure 5 highlights this idea: while point estimates remain the same with very little variation, a low number of clusters for standard error inflates them and therefore leads to loss of statistical power rather than absence of an effect. This limitation and its potential solutions are further described in subsection 5.2.

5 Discussion

5.1 Seasonality and Climate Risk in Housing Prices

The regression results reveal several things about the seasonality and capitalization of climate risk at the Jersey Shore.

First, the FEMA flood zone does not have any significant effects on the sale price. Therefore, this study does not find any statistically significant association between flood zones and sales price, nor any interaction between flood zones and seasonality. This could be explained by a variety of factors, including local beliefs and information accessibility, that have been shown to affect the capitalization of climate risk into housing prices (Baldauf et al., 2020; Chen, 2025; Hino & Burke, 2020).

Second, there is a weak positive association between seasonality and sales price, indicating that the more seasonal, or, in other words, tourist-oriented, the parcel's area becomes, the higher the sale price. The speculative explanation for it is that when areas become seasonal, meaning that other parcels within the census tract are seasonal, the sale transactions that occur are oriented towards similar utilization of the parcel. In other words, as market actors see that their parcel is located in an area with high seasonality and that demand for housing is therefore high, they increase the selling price, as they understand that other market actors are eager to invest in such property due to its potential to generate profits.

Third, we find the statistically significant association between distance to the shore and sale price. The negative coefficient in model 4 reflects a well-documented coastal amenity premium.

Being close to the ocean, having ocean views, and proximity to the beach are amenities that market actors value, increasing the sale price of the parcel. Notably, this premium exists despite properties closer to the ocean facing higher climate risks from flooding, natural hazards, and long-term sea level rise. This finding is consistent with literature documenting overvaluation of flood-exposed coastal properties (Gourevitch et al., 2023; Murfin & Spiegel, 2020).

Fourth, and most importantly, in light of the study’s research question, we find that the interaction between distance and the percent of seasonal units is significant across all model specifications. A negative coefficient in models 2 and 3 indicates that properties located closer to the ocean, in areas with a higher share of seasonal housing units, are valued higher, and this effect decays with distance. Specifically, the full marginal effect of a unit increase in Pct. Seasonal on log sale price is $0.188 - 0.196 \times \log(\text{Distance})$, meaning the seasonality premium is positive and largest for properties close to shore, diminishes with distance, and reverses for properties far inland, at approximately 2.6 miles. More broadly, it means that the price premium for seasonality is not uniform across the Jersey Shore. This finding is economically intuitive: shore access matters for seasonal units, as buyers interested in a second home or short-term rental want to be as close to the beach as possible.

5.2 Limitations and Future Research

Results of this study should be interpreted with several limitations in mind. These limitations led to potential avenues for future work, discussed below.

First, as has been previously discussed in the subsection 3.4.2, one of the independent variables of interest, percent of the seasonal housing units, is collected at the level (census tract) other than the other variables, which creates a modifiable areal unit problem (MAUP) (Lee et al., 2016). As census tracts are spatial units that have nothing to do with the housing market, during the data generation process, housing units that share the same seasonality character can be assigned different values due to being in different census tracts. Moreover, MAUP can affect regression results on the respective variable, as changing the unit of data collection, for instance, from census tract to census block or ZIP code, will respectively change regression coefficients, undermining the robustness of study findings.

One potential solution to this problem in the current study is to use an alternative proxy measure, for instance, through the owner’s address. MOD-IV data provide information on the owner of the parcel address, so if we compare it with the parcel’s physical address, we can generate a variable indicating whether the owner lives in the parcel or in another area of the state or another state. Quick validation of this variable yielded a Pearson correlation coefficient of 0.7 when compared with the aggregated owner proxy at the census level seasonality. However, it is important to note that such a proxy would not capture the distinction between renters and seasonal residents. Therefore, additional efforts are required to design a good seasonality proxy.

Second, there is a limitation in the clustered standard errors design of the hedonic regression, discussed in section 4. While reporting clustered standard errors is required given the study’s

sample size, there is a small number of clusters issue that inflates standard errors, leading to a loss of statistical power and non-significant coefficients. A potential solution to that limitation in future research might include clustering standard errors at a larger scale, such as state municipalities.

This also leads to another limitation. While introducing time-fixed effects has been shown to improve model fit and account for unobserved heterogeneity that affects sales prices, introducing entity-fixed effects at the county level does not yield a significant improvement in the model. Similar to the previous limitation, future research should consider implementing fixed effects at a larger scale or moving towards other approaches that account for spatial dependence, such as spatial error and lag models.

Lastly, while our control variables have proved to be robust, hedonic models of property values usually benefit from the introduction of a broader range of building characteristics controls. MOD-IV data provides that information through the building description variable, though parsing it into separate variables was challenging during data cleaning. Therefore, future studies should consider ways to generate such variables to incorporate better building characteristics controls into hedonic models.

6 Conclusion

This study has examined the interaction of seasonality and climate risk capitalization in the housing market at the Jersey Shore. Using tax accessor MOD-IV data on parcel sales, as well as FEMA flood maps data and spatial analysis tools, I generate pooled cross-sectional data for 10 years of sales transactions. Then, a hedonic regression analysis is performed with year and county fixed effects on log-transformed sales price against flood zones, distance to shore, share of seasonal housing units, and interaction terms between those variables, along with specified control variables.

The main finding of this study reveals that the flood zone does not have a robust, significant effect on the sale price. However, distance to the shore and the share of seasonal units do affect prices: properties located close to the shore receive a price premium compared to those further inland. Moreover, the effect of seasonality is positive: a higher share of housing units in a census tract is associated with a higher price. However, this premium is largely conditioned on proximity to the shore. This effect decreases as the distance to shore increases, meaning that properties located further inland in high-seasonality areas would receive a discount rather than a premium.

6.1 Policy Implications

These findings have a few important policy implications. The seasonality and distance-to-shore premium in housing prices are not independent; rather, they are concentrated in areas close to the shore. Properties in high-share of seasonal housing areas have higher market prices as market actors are interested in second homes or short-term rentals in proximity to the beach. However, these properties also have the highest climate risk. While in accordance with scholarship (Gourevitch et al., 2023), the current coastal housing market can be speculatively called overvalued,

there is a possibility that various events, such as publication of new FEMA maps, natural hazards, or disclosure laws, might lead to the market correction, and lower-income and year-round communities might be at the greater risk due to the fact that these properties are their main assets.

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Data Availability

All the data used in this paper are publicly available online.

AI Usage

This project has utilized Claude Code, integrated into the programming environment, for debugging and code adjustment. Prompts consisted of "debug [this] function" or "fix [this] issue in the code", "adjust [this] to be [this]". No parts of the final paper were generated using AI. Grammarly was used for proofreading.

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